

SASTRA DEEMED UNIVERSITY
(A University under section 3 of the UGC Act, 1956)

End Semester Examinations

Nov 2025

Course Code: CIV102

Course: ENGINEERING MECHANICS

QP No. :S1712-1

Duration: 3 hours

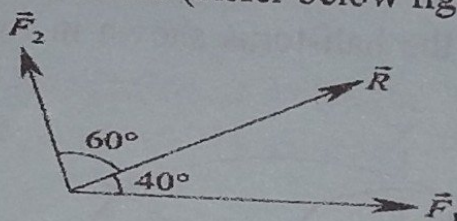
Max. Marks:100

PART - A

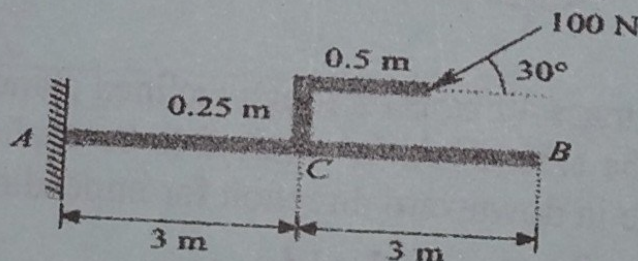
Answer all the questions

10 x 2 = 20 Marks

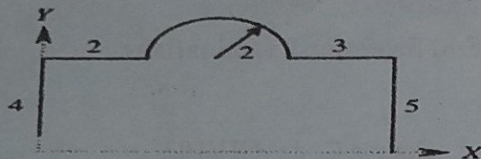
1. The resultant of two forces is 400 N. If the forces are inclined at 40° and 60° with the resultant, one on either side, calculate the magnitude of the two forces (Refer below figure).



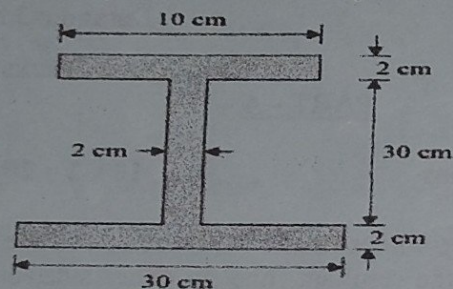
2. A force 100 N is passing through a line AB with coordinates A (1,2,3) and B (5,8,12). Find the force as a vector.
3. Compute the moment of a 100 N force applied on a cantilever beam as shown in figure about the fixed end A.



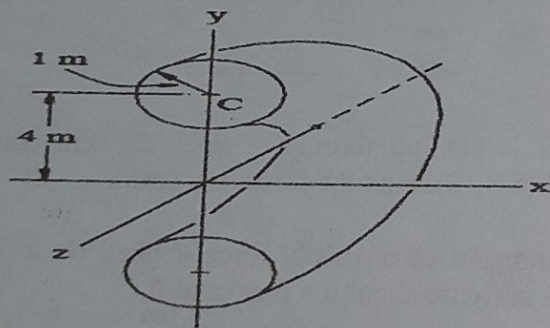
4. Find the centroid of the bent wire shown in figure. All dimensions are in cm.



5. The cross section of a section is shown in figure. Find the radius of gyration about the symmetric centroidal y-axis.



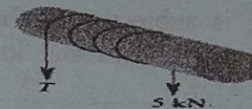
6. Determine the area of the half-torus shown in figure by Pappus-Guldinus theorem.



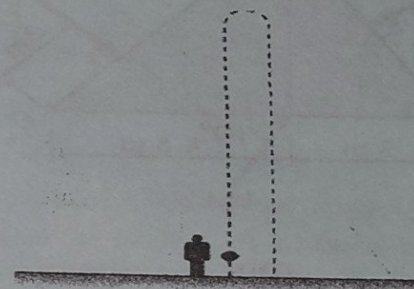
7. A wooden block of mass 40 kg is on rough inclined plane as shown in figure. What is the external force required to be applied parallel to the inclined plane in downward direction for impending motion? Take $\mu = 0.4$.

$\theta = 15^\circ \rightarrow$ the horizontal

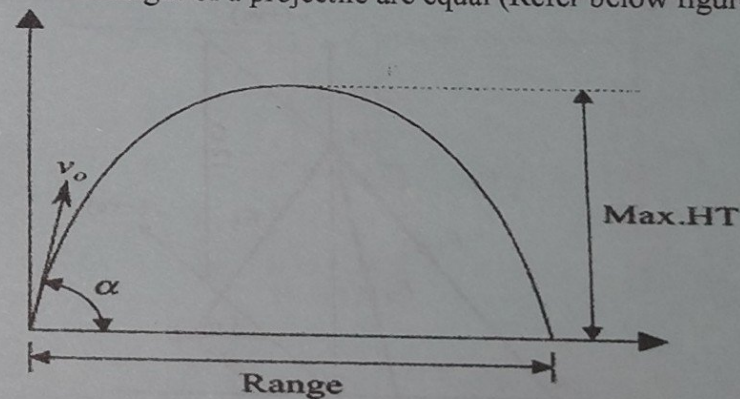
8. A rope is wrapped three and a half times around a cylinder. Determine the force T exerted on the free end of the rope that is required to just support a 5 kN weight. The coefficient of friction between the rope and the cylinder is 0.2.



9. A ball is thrown vertically upwards from the ground with an initial velocity of 20 m/s. Determine (i) the maximum height reached by the ball, (ii) the time taken to reach the maximum height, and (iii) the total time of flight.



10. Find the angle of projection at which the horizontal range and the maximum height of a projectile are equal (Refer below figure).

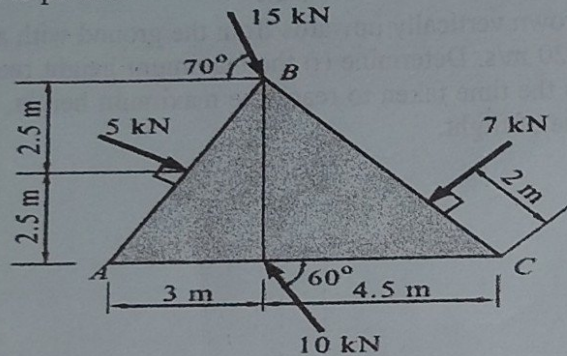


PART - B

Answer all the questions

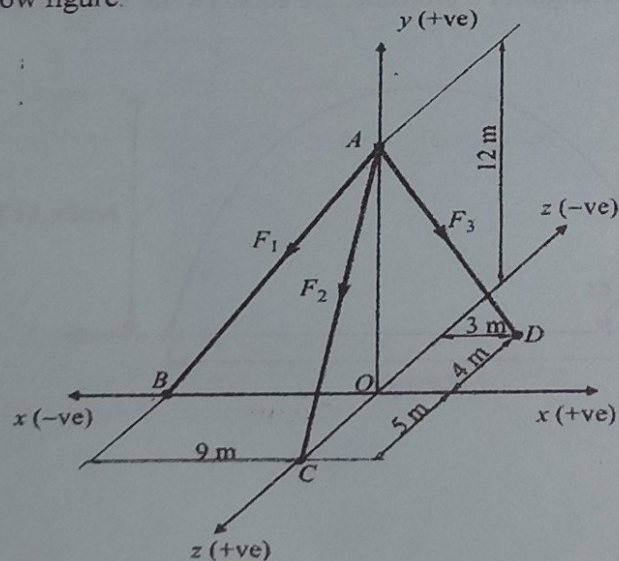
4 x 15 = 60 Marks

11. A triangular plate ABC is subjected to four coplanar forces as shown in figure. Find the resultant and locate its position with respect to point A.

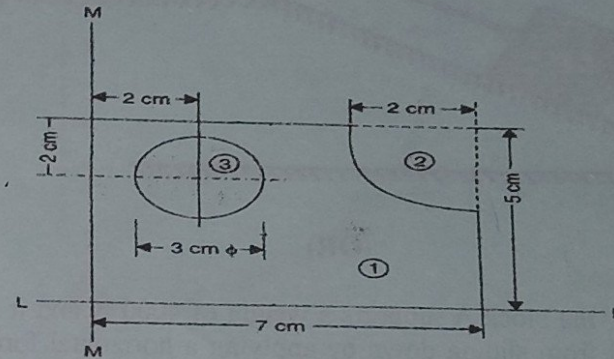


(OR)

12. The resultant of the three concurrent space forces at A is $R = (-788 \text{ j}) \text{ N}$. Find the magnitude of F_1 , F_2 and F_3 forces as shown in the below figure.

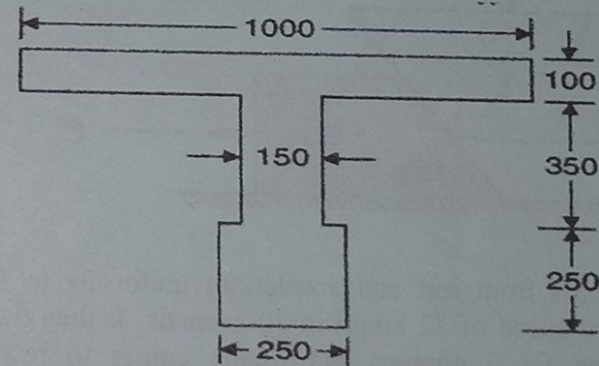


13. Determine the centroid of the plane figure shown below.

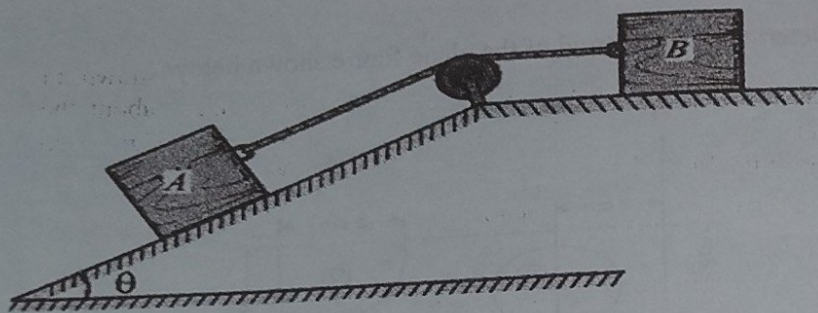


(OR)

14. The cross-section of a prestressed concrete beam is shown in figure. Calculate the moment of inertia of this section about the centroidal axes parallel to and perpendicular to top edge. Also determine the radii of gyration. All dimensions are in mm.

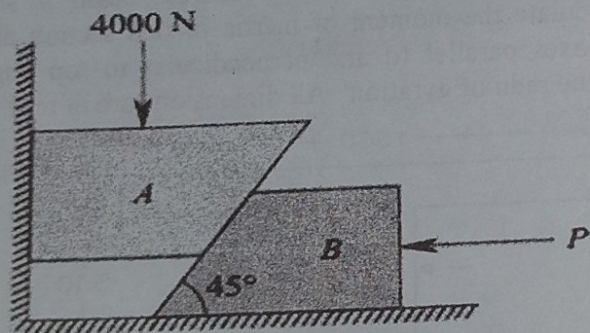


15. Find the value of θ if the blocks A and B shown in figure have impending motion. Given block A = 20 kg, block B = 20 kg, $\mu_A = \mu_B = 0.25$.



(OR)

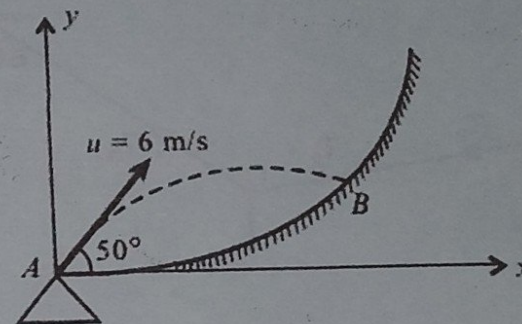
16. In Figure, the block A supports a weight of 4000 N and it is to be prevented from sliding down by applying a horizontal force P on the block B. If the coefficient of friction at all surfaces of contact is 0.2. Determine the smallest force P required to maintain equilibrium. Neglect the weights of the blocks.



17. a) A car starts from rest and accelerates uniformly to reach a maximum speed of 72 kmph in 30 seconds. It then travels at this speed for 3 minutes and finally comes to rest in 45 seconds. Determine (i) the acceleration and deceleration of the car, (ii) the total distance travelled during this time, (iii) average velocity in this time, and (iv) average acceleration in this time. (10)

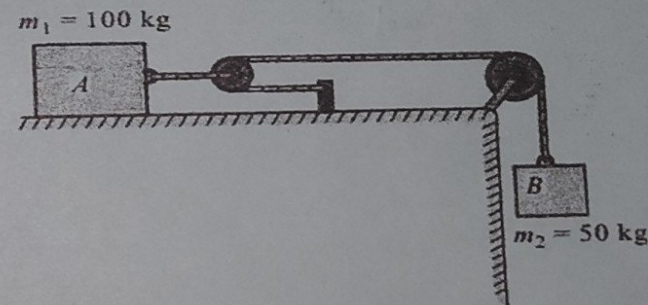
- b) The water sprinkler positioned at the base of a hill releases a stream of water with a velocity of 6 m/s, as shown in Figure.

Determine the point B (x , y) where the water particles strike the ground on the hill. Assume that the hill is defined by the equation $y = 0.2x^2$ m and neglect the size of the sprinkler. (5)



(OR)

18. Two blocks, shown in figure, start from rest. If the cord is inextensible, friction and inertia of pulley are negligible, calculate acceleration of each block and tension in each cord. Consider coefficient of friction as 0.25.



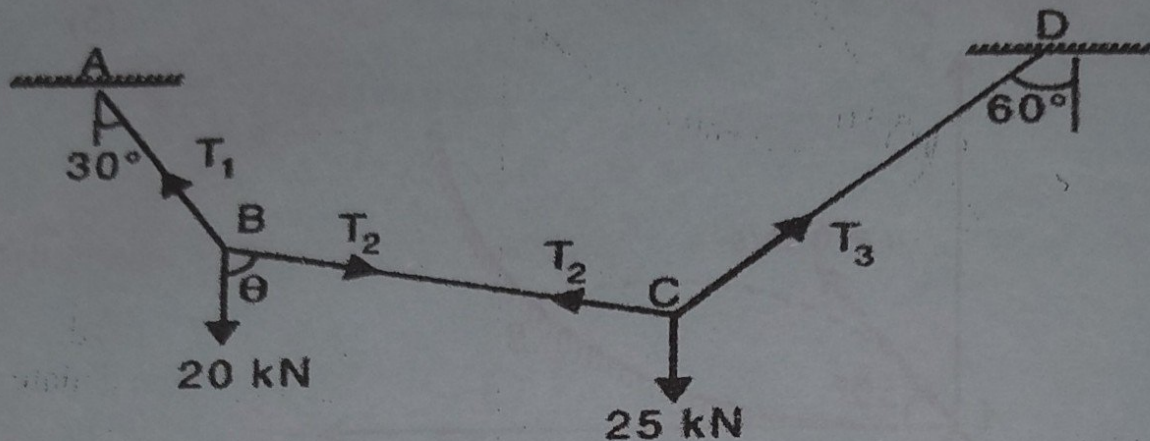
PART - C

Answer the following

1 x 20 = 20 Marks

19. a) A wire is fixed at two points A and D as shown in figure. Two weights 20 kN and 25 kN are supported at B and C, respectively. When equilibrium is reached it is found that inclination of AB is 30° and that of CD is 60° to the vertical.

Determine the tension in the segments AB, BC and CD of the rope and also the inclination of BC to the vertical.
(10)



- b) The acceleration of a particle is defined by the relation $a = 25 - 3x^2 \text{ mm/s}^2$. The particle starts with no initial velocity at the position $x = 0$, determine (i) the velocity when $x = 2 \text{ mm}$, (ii) the position when velocity is again zero and (iii) position where the velocity is maximum and the corresponding maximum velocity.
(10)
